

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours**Maximum Marks:35**

Q.1 Write a C program to simulate Banker's algorithm for the purpose of deadlock avoidance. Consider the following snapshot of system, A, B, C and D is the resource type.

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- a) Calculate and display the content of need matrix?
 b) Is the system in safe state? If display the safe sequence.

[15 Marks]

Q.2 Write a simulation program for disk scheduling using SSTF algorithm.
 Accept total number of disk blocks, disk request string, and current head position from the user. Display the list of requests in the order in which it is served. Also display the total number of head moments.
 98, 183, 37, 122, 14, 124, 65
Start Head Position: 53

[15 Marks]

Q.3. Oral/Viva [05 Marks]